

QA REVIEW REPORT: MOBILE APPLICATION

AcePharm — iOS & Android Flutter Companion App (v0.1.2) | QA Officer: Hafiz Saad | Developer: Numan Latif | Date: 18/09/2026

Review Summary

Overall Result	FAIL — not ready for App Store / Play Store release
Review Date	18/09/2026
Project	AcePharm — Cross-Platform Mobile App (Build under test: v0.1.2 / Bundle: com.acepharm.app)
Lead Developer	Numan Latif
QA Reviewer	Hafiz Saad (Senior QA Officer, Takween Digital)

Category / Audit Domain	Audit Status
Project Setup & Platform Bundles	Pass
Authentication & 5-Step Mobile Onboarding	Pass
Question Engine & Answering Interaction Rules	Partial
Clinical Explanations & RAG 'Ask Ace'	Fail
Offline Resilience & Hospital Ward Sync	Fail
Progress Analytics & Visual Topic Map	Partial
Account Management, Deletion & Store Compliance	Fail

Notes to Developer

This QA audit evaluated the native Flutter mobile application (build v0.1.2) on physical test devices (iPhone 14 iOS 17.4 and Google Pixel 7 Android 14) following the pre-release manual QA plan. Automated unit testing is robust (116 passing tests), Flutter ThemeData faithfully applies the AcePharm design tokens (#4F46E5, Geist fonts, 44pt touch targets), and onboarding step navigation is fluid.

However, the mobile build is rated **FAIL** due to three critical architectural flaws that compromise core product invariants. Most critically, **MOB-01** demonstrates that submitting question attempts while in Airplane Mode (simulating hospital ward basements) drops local attempt timestamps and fails to sync when network resumes, corrupting calibration metrics. Second, **MOB-02** shows Ask Ace generating unbounded hallucinated drug interactions when queried on out-of-domain substances rather than executing the mandated graceful refusal. Third, **MOB-03** violates Apple App Store Guideline 5.1.1: the Account Deletion workflow triggers a client-side logout without permanently deleting the user record from the PostgreSQL database.

Issues to Correct — Critical & Fail

[CRITICAL] MOB-01 — Offline Resilience — Hospital Ward Queueing — Offline question attempts drop timestamps and fail to sync upon reconnection

Issue: When completing revision in Airplane Mode, submitted question attempts are cached in Hive but their local execution timestamps are lost. Upon disabling Airplane Mode, the background sync service fails silently on retry, resulting in lost practice records and corrupted confidence calibration.

Fix Required: Re-architect the offline synchronization engine: persist attempt payloads (question_id, selected_option, confidence, duration_ms, timestamp_iso) in SQLite/Hive with exponential backoff retry. Ensure sync completes automatically upon connectivity restoration without user intervention.

QA Note: Violates Product Invariant #6 (Hospital ward connectivity resilience). Zero attempt loss is mandatory.

[CRITICAL] MOB-02 — Ask Ace AI Tutor — Guardrails & Grounding — Ask Ace hallucinates responses for unindexed drugs instead of graceful refusal

Issue: When querying Ask Ace regarding an unindexed investigational oncology agent, the mobile client generated a speculative 3-paragraph answer with synthetic citations rather than displaying the required neutral refusal message ('This topic is outside AcePharm's reviewed curriculum').

Fix Required: Enforce pgvector distance threshold gating in the mobile API bridge: if semantic similarity score falls below 0.78, force the pre-authored refusal response. Disallow unbounded model completion.

QA Note: High clinical risk. Students on hospital placements rely on Ace for examination study; hallucinations destroy platform credibility.

[CRITICAL] MOB-03 — Settings & Compliance — Account Deletion Flow — *Account Deletion only clears local tokens; user record persists on server*

Issue: Executing Settings → Account → Delete Account prompts for password and redirects to login. However, testing verified the user record remains active in PostgreSQL, and re-logging in with identical credentials succeeds immediately. This directly violates Apple App Store Review Guideline 5.1.1(v).

Fix Required: Wire the native deletion confirmation to the backend GDPR deletion endpoint (DELETE /api/v1/user/account), wiping authentication credentials and setting status to 'pending_deletion'.

QA Note: App Store review rejection guaranteed if account deletion does not actually delete the server-side account.

[FAIL] MOB-04 — Question Engine — Timed Exam Mode — *Timed Exam mode reveals immediate option rationales before submission*

Issue: In Timed Exam Mode, selecting an option inadvertently expands the explanation accordion, revealing the correct answer and option rationales before the user taps 'Submit Answer'.

Fix Required: Strictly decouple option selection from explanation visibility. In Timed Mode, suppress all rationales and correct-answer styling until the session is formally submitted.

QA Note: Defeats the purpose of timed exam practice.

[FAIL] MOB-05 — Progress Analytics — Metric Presentation — *Mobile progress screen collapses 5 metrics into a single weighted score*

Issue: The mobile progress dashboard displays a consolidated 'Mastery Score (74%)', directly violating Non-Negotiable Rule #1 ('Five metrics kept visibly distinct, never collapsed into one score') and Rule #19 ('Avoid the word mastered').

Fix Required: De-collapse the score into the five distinct metrics: First-Attempt Accuracy, Practice Accuracy, Repeat Accuracy, Confidence Calibration, and Coverage. Eliminate 'Mastery' terminology.

QA Note: Violates core AcePharm educational philosophy.

[FAIL] MOB-06 — Monetization — In-App Purchase Compliance — *Upgrade button directly opens external Stripe web checkout from within app*

Issue: Tapping 'Upgrade to Full Access' invokes a webview pointing to Stripe Checkout. Apple Guideline 3.1.1 prohibits linking to external web payment mechanisms for digital content unlocked inside the iOS app without implementing Apple StoreKit In-App Purchases.

Fix Required: Adopt the compliant 'Reader App / Status Display' approach: show active subscription status if purchased on web, but replace the direct payment link with approved neutral copy: 'Subscriptions can be managed via your web account settings'.

QA Note: Guaranteed iOS App Store rejection under guideline 3.1.1.

Issues to Correct — Partial & Minor

ID	Section	Issue	Required Correction
MOB-07	Calculations	Geist Mono numeric alignment breaks on small Android viewports (360px).	Wrap calculations and formula blocks in horizontally scrolling containers or adjust font scaling.
MOB-08	Session Builder	High-risk category chips do not display the amber clinical warning icon.	Add semantic amber alert badge alongside high-risk drug categories.
MOB-09	Topic Map	Six-stage status taxonomy shows 'Needs Review' instead of spec'd 'Due for Review'.	Correct status taxonomy label to match Developer Brief Section 4 exactly.
MOB-10	UI Polish	Explanation accordion tap target measures 38×38pt on compact phones.	Increase hit-test padding to guarantee minimum 44×44pt touch target.

Overall Impression & Strategic Assessment

Strongest Area	Clean Flutter architecture, high-fidelity design token translation (#4F46E5, typography), smooth onboarding flow, and rapid 60fps question scrolling performance.
Weakest Area	Offline network resilience, out-of-domain AI grounding safety, and Apple App Store compliance (account deletion and external payment links).
Red Flags	MOB-01 (silent attempt loss in offline hospital mode); MOB-02 (Ask Ace clinical hallucination); MOB-03 (fake account deletion violating Apple Guideline 5.1.1); MOB-06 (Stripe external link violating Guideline 3.1.1).
Recommendation	Do not submit to App Store or Play Store production tracks. Suggested remediation roadmap: (1) Fix server-side Account Deletion; (2) Remove Stripe external payment link; (3) Implement robust offline SQLite attempt queueing; (4) Enforce pgvector threshold refusal in Ask Ace; (5) De-collapse progress metrics into the five discrete indicators.

Sign-Off Status

QA Officer	Hafiz Saad (Senior QA Officer)
Developer	Numan Latif (Mobile Engineer)
Date Resolved	Pending Re-audit
Formal Verdict	REJECTED (FAIL) — Resolve Critical App Store blockers prior to TestFlight

A formal re-audit will be conducted on build v0.1.3 once offline queueing and Apple compliance requirements are fully satisfied.